Quality Control (QC) and Quality Assurance (QA) Procedures for Meteorological Data from Automatic Weather Stations

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Abstract **- Meteorological Data is received from automatic weather stations in a raw format. Quality control (QC) and Quality Assurance (QA) are vital for effectively utilizing meteorological information, ensuring reliability and precision for diverse applications. High-quality data is indispensable for scientists, engineers, and decision-makers who rely on accurate meteorological insights for improved weather prediction, research, planning, and decision-making processes. This study presents and puts into practice extensive quality check and assurance techniques to assess meteorological data from Automatic Weather Station (AWS) networks that are adequately maintained year-round and have their performance cross-checked with traveling standards. These procedures include validation of data record structures, range and limit checks, temporal and internal consistency assessments, persistence tests, and spatial consistency evaluations. This approach identifies multiple error types and facilitates corrective actions such as preventive or corrective maintenance, bias corrections, calibration, sensor replacements, and data exclusions. This study investigated the QA/QC of data from 13 AWSs collected between January 2023 to October 2024 from the Pune Districts of Maharashtra, India. The performance evaluation of automatic weather stations (AWS) through range and consistency checks demonstrates high reliability, with average data validity exceeding 95% across key parameters, indicating their effectiveness for accurate meteorological monitoring.**

Keywords - Automatic Weather Stations (AWS), Quality Control (QC), Quality Assurance (QA), Weather Data Processing, Data Quality Checks, Calibration, Error Detection.

I. INTRODUCTION

Quality Control (QC) ensures the accuracy, consistency, and reliability of meteorological data collected from various observation stations. In meteorology, reliable data is indispensable for various applications, including weather forecasting, climate studies, and environmental monitoring [1]. The primary objective of QC is to detect and address errors that may arise during the data collection process, ensuring that the data provided to end-users is accurate and valid. By minimizing these errors, QC enhances the quality of data, which is fundamental for making informed decisions, whether in scientific research, engineering, or policy-making [2]. QC is typically initiated by checking the feasibility of the equipment conditions at observation stations. This involves verifying the operational status of the equipment, ensuring that it is correctly calibrated and functioning as intended [3,4]. The QC process then examines the data collected by these instruments, whether manually or automatically operated. Manual data collection methods often involve direct observation and recording by individuals, while automated systems, such as Automatic Weather Stations (AWS), continuously monitor and record meteorological parameters at predefined intervals [5]. The need for automated QC becomes evident given the data generated from these systems, particularly in high-frequency observations. As the volume of meteorological data grows exponentially, especially with the increasing number of observation stations and the frequency of measurements (e.g., hourly, every 15 minutes, or even minute-by-minute), performing QC manually becomes increasingly complex and inefficient. Therefore, automating the QC process within the data collection systems is essential to maintaining the data's integrity and managing the large influx of real-time data effectively [6,7]. Automated QC methods are particularly beneficial in dealing with massive high-frequency data generated by systems like AWS.

The World Meteorological Organization (WMO) recognizes the importance of automated QC in managing large datasets. It ensures the methods can provide reliable results despite the potential challenges of high-frequency data [3,4]. The two primary methods include range checks and consistency checks. Range checks ensure the collected data falls within each monitored parameter's expected or acceptable climatological limits. For example, temperature readings that exceed or fall below the typical range for a given location may indicate an equipment malfunction or a data error. On the other hand, consistency checks evaluate the relationship between different parameters, ensuring that they remain consistent [8,9]. For instance, air pressure and temperature are often correlated, and any inconsistency between these parameters could suggest faulty data or sensor issues. In addition to range and consistency checks, WMO also recommends three other methods based on statistical tests: the flat line test, step check, and spike test. The step check evaluates sudden, significant changes in a parameter, comparing the current value with the previous observation [10]. The India Meteorological Department (IMD) is the nodal weather forecasting and monitoring agency. It maintains a network of strategically placed networks of AWS / ARG / AGRO-AWS across the country. IMD Central Servers for AWS data receptions have implemented these QC methods into its automated systems for monitoring meteorological data, including range, consistency, and step checks. These checks are applied to the data collected from various AWS networks across India, which include parameters such as rainfall, air pressure, air temperature, humidity, solar radiation, wind speed, etc. These stations are regularly examined for data drifts and are kept in good condition. Values are routinely compared with portable reference standards/traveling standards [11-13]. This study aims to evaluate the effectiveness of preventive and corrective maintenance and calibration practices, which leads to better QA/QC data from AWS stations. QC methods ensure the data's validity and propose improvements based on the findings. In this study, AWS station data collected from the Pune Districts of Maharashtra, India, were investigated, and the performance of the current automated QC methods was validated. The goal is to assess the reliability of the data after applying QC procedures and to identify areas where the methods can be refined. Implementing automated quality assurance and quality checks in AWS systems is vital for ensuring the reliability and accuracy of meteorological data. As demand for real-time weather data grows, automated QC methods will become even more critical [14]. By enhancing these methods and addressing the challenges associated with variable weather conditions, the effectiveness of AWS systems can be improved, ultimately supporting better weather predictions, climate modeling, and decision-making in both operational and research domains [15-17]. This study contributes to the ongoing efforts to optimize the QA/QC process for AWS data and improve the overall quality of meteorological observations.

II. LITERATURE REVIEW

A. Automatic Weather Station

An Automatic Weather Station (AWS) system is designed to measure, record, and transmit meteorological data autonomously. It comprises various components, including sensors, data loggers, communication systems, power supply units, displays, and other supporting equipment. The AWS is equipped with five primary sensors: The Air Temperature and Relative Humidity (ATRH) sensor, an air pressure sensor, an ultrasonic wind sensor for measuring wind speed and direction, a Pyranometer for monitoring solar radiation, and a Tipping Bucket Rain Gauge (TBRG) for capturing rainfall data. These sensors collect weather parameters, which the data logger processes [16]. The processed data is transmitted via File Transfer Protocol (FTP) using a GPRS modem or satellite communication and uploaded to web portals using the Hypertext Transfer Protocol (HTTP) method. This entire process is automated and operates in real time, with data updates typically occurring every 15 minutes. Acquiring high-quality data from an AWS station presents several challenges. These may include missing values, errors in readings, sudden jumps in data caused by power disruptions, issues with sensor electronics, sensor aging, drift in sensor calibration, exposure conditions, and transmission failures. Ensuring quality assurance and performing quality checks are essential [17]. Table I presents the specifications of the AWS Sensors.

TABLE I. SENSOR SPECIFICATIONS

S no.	Variable	Accuracy	Resolution	Range
1	Air Pressure (hPa)	0.1	0.1	600-1100
2	Relative Humidity (%) Capacitive Solid state	2	1	0 -100
3	Air Temperature (°C) PT 100 RTD	0.2	0.1	-40 to 60
4	Wind Speed (m/s) ultrasonic wind sensor	0.5	0.1	0 to 75
5	Wind Direction (degree) ultrasonic wind sensor	5	1	0-359
6	Precipitation(mm) with tipping bucket rain gauge	-	0.5	-
7	Solar Radiation (nm)	5%	-	350-1100

B. Quality Control & Assurance Method

1) *Regular calibration* with Traveling standards at the field,[6] if necessary, replace sensors

$$Mean\ Bias\ Error\ (MBE) = \frac{1}{n}\sum_{i=1}^{n}(S_i - R_i) \quad (1)$$

$$Relative\ mean\ bias\ Error\ (RMBE) = \frac{\sum_{i=1}^{n}(S_i - R_i)}{\sum_{i=1}^{n} R_i} X100\% \quad (2)$$

$$Precision = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(S_i - R_i - MBE)^2} \quad (3)$$

$$Relative\ precision = \frac{Precision}{\frac{1}{n}\sum_{i=1}^{n} R_i} X\ 100\% \quad (4)$$

Where S = Sensor Value, R = Reference Sensor Value and n = number of samples.

2) *Range Check*

An essential quality control (QC) stage in data validation is range checking, which compares values to predetermined lower and upper ranges based on the local climate. This technique measures climatic variables like rainfall, wind speed, humidity, air temperature, and air pressure. Statistical analyses of large historical data series determine the limit values [6,11]. Table II lists the predefined boundaries of the AWS Center system, which are generally applicable to all AWS sites in India.

TABLE II. RANGE CHECK LIMIT VALUE

S no.	Parameter	Lower Limit	Upper Limit
1	Air Pressure (hPa)	600	1100
2	Relative Humidity (%)	0	100
3	Air Temperature (C)	-40	60

4	Dew Point (C)	10	35
5	Wind Speed at 10 m(knots)	0	75
6	Wind Direction at 10m(degree)	0	360

3) Consistency check

A consistency check identifies discrepancies between a parameter and related parameters marked as suspicious. This test can measure several meteorological factors, including relative humidity, wind direction, average speed and maximum speed, sun radiation, and air temperature (including average, maximum, and minimum temperatures) [3]. Internal consistency of data is based on the relation between two parameters.

The wind is erroneous and flagged in the following cases:

wind direction = 00 and wind speed ≠ 00

wind direction ≠ 00 and wind speed = 00

wind gust (speed)≤ wind speed

The temperature information is erroneous and flagged in the following case:

dew point temperature > air temperature

The precipitation information is erroneous and flagged in the following cases:

Precipitation was recorded, but no sudden change in temperature(decrease) and humidity(increase) were observed, indicating possible issues with the (TBRG) rainfall sensor reporting false precipitation. Typically, precipitation is accompanied by a decrease in temperature and an increase in humidity.

4) Step Check

A step check assesses how much a parameter's value can fluctuate over a given time, like every hour or every fifteen minutes. Usually, non-meteorological variables sometimes are responsible for variations surpassing the tolerance level. The threshold, also known as the tolerance limit, can be dynamic, depending on the results of tests conducted in real-time or static, based on a statistical analysis of long-term historical data. The AWS Center system uses a dynamic threshold determined by the standard deviation approach [1,6]. The equation below describes the criteria for judging whether data passes the test for a 15-minute interval. The current instantaneous value fails to check if the following condition is not fulfilled.

$$|Vi - Vi\text{-}1| + |Vi - Vi\text{+}1| \leq 4\sigma V \quad (5)$$

Where Vi is the current value of the parameter,
Vi-1 is the previous value of the parameter,
Vi+1 is the next value of the parameter,
σV is the standard deviation

Equation (5) exploration in terms of temperature, if V_i represents the current temperature values, V_{i-1} the previous temperature, and V_{i+1} the next subsequent temperature, the condition ensures that the temperature does not change abruptly in an unrealistic manner [16,17]. The sum of absolute difference, $|V_i - V_{i-1}|$ and $|V_i - V_{i+1}|$, must be less than or equal to 4 times the standard deviation (4σV). If this condition is not satisfied, the temperature readings are inconsistent or potentially erroneous, as abrupt changes are unlikely under normal meteorological conditions.

5) Persistence test

Possible limits of the minimum required variability Air temperature: 0.1°C over the past 60 minutes can be as follows [6]. Dew point temperature:0.1°C over the past 60 minutes. Relative humidity: 1% over the past 60 minutes (Only if the measurement relative humidity is less than 95% to consider uncertainty). Wind direction: 10 degrees over the past 60 minutes (Only if the 10-minute average wind speed during the period is more significant than 0.1 m/s). Wind speed: 0.5 ms-1 over the past 60 minutes (Only if the 10-minute average wind speed during the period is more significant than 0.1 m/s)

6) Spatial Consistency Check:

A spatial quality check algorithm for Automatic Weather Station (AWS) data involves comparing the observations from one station to neighboring stations to assess the reliability of measurements [11,12]. Spatial quality checks are beneficial for identifying outliers or errors in data due to sensor malfunctions or transmission issues.

7) Quality Assurance (QA) of AWS/ARG Station Data:

QA involves proactive measures to maintain data integrity before deployment, including selecting high-precision sensors that comply with WMO standards, factory calibration, and periodic recalibration. Proper site selection, exposure conditions, installation, deployment, sensor alignment, and stable anchoring [13]. Redundant stations help cross-check data accuracy. Network reliability is ensured through stable power, communication, and backup systems. Maintaining metadata and senor logs, including field history of preventive and corrective maintenance records. and software updates, further enhancing data accuracy and operational efficiency. Regular training programs for manpower ensure capacity building, equipping personnel with skills needed for effective AWS operation, maintenance, and data quality assurance [14,15].

III. DATA AND METHODOLOGY

A. Dataset

Quality control (QC) was performed on data from 13 AWS stations in Pune District, Maharashtra, India, as listed in Table III. These stations generate various meteorological parameters, including RAIN_HOURLY(mm), RAIN_DAILY(mm), TEMP(°C), TEMP_MIN(°C), TEMP_MAX(°C), TEMP_MIN_MAX_DAY(°C), DEW_POINT (°C), RH (%), RH_MIN_MAX_DAY (%), WIND_DIR_10m(°), WIND_SPEED_10m(kt), WIND_MAX/GUST_10m(kt), WIND_DIR_3m(°), WIND_SPEED_3m (kt), WIND_MAX/GUST_3m (kt), SLP (hPa), MSLP (hPa), and SUNSHINE (HH:MM). The QC process was applied to data collected from January 2023 to October 2024. It involved range checks, consistency checks, and step checks, each designed to detect and address errors in the data. Such errors could arise from various sources, including failures in electronic components, sensor malfunctions, power interruptions, or a combination of these factors. Additionally, TDMA encoding/decoding issues during data transmission could contribute to inaccuracies—approximately 89% of data available from the AWS stations.

TABLE III. AWS STATIONS IN PUNE, INDIA

S no.	Station	Latitude (° N)	Longitude (°E)	Elevation(m)
1	JUNNAR	19.2092	73.8725	619
2	SHIVAJINAGAR	18.5386	73.842	559
3	LAVASA	18.4144	73.5069	687.3
4	DAPODI	18.6032	73.8541	563.2
5	HADAPSAR	18.4659	73.9244	618.7
6	LONAVALA	18.724	73.3697	620.8
7	DAUND	18.5056	74.3304	554
8	LONIKALBHOR	18.4697	74.0013	563
9	NARAYANGOAN	19.1003	73.9655	694.5
10	BARAMATI	18.153	74.5003	569.7
11	PASHAN	18.5167	73.85	577
12	RAJGURUNAGAR	18.841	73.884	598.1
13	TALEGAON	18.722	73.6632	635.8

B. *Methodology*

The steps for performing AWS data quality control (QC) are as follows: First, data is collected from 13 AWS stations in Pune, Maharashtra, India. Duplicate timestamps are checked and removed for each site to ensure data accuracy. The data is then organized into a continuous time series from 1st January 2023 at 00:00 UTC to 31st October 2024 at 23:45 UTC, with 15-minute intervals. Sites with minute-level data are adjusted to 15-minute intervals using appropriate aggregation methods for parameters such as air pressure, humidity, wind speed, and temperature. A range check compares the data with predetermined limit values, excluding any data that falls outside the acceptable range from consistency and step checks. Consistency checks are then performed by comparing parameters like air temperature, solar radiation, and wind, assessing their relationships, and counting the number of consistent data pairs. Data that passes the range check for air pressure, humidity, and average air temperature is then prepared for step checking by arranging it into a continuous time series. During step check execution, the standard deviation for the last six data points is calculated, and the differences between the tested data and previous points are evaluated. Data passes the step check if all differences remain within the standard deviation limits. Additionally, a persistency check is performed to ensure minimal parameter variation over time, confirming that fluctuations remain within acceptable limits (typically within 1-hour or 15-minute intervals) to ensure data stability.

IV. RESULT AND DISCUSSION

Traveling standards or portable reference standards serve a significant role in correcting sensor biases at AWS/ARG field sites. They guarantee that AWS sensors will retain accuracy over time by offering a trustworthy, calibrated standard against which to compare. Discrepancies brought on by aging, environmental influences, or sensor drift can be reduced by routinely employing these standards for calibration checks. This enhances the general quality and dependability of the AWS network by preserving consistent and trustworthy data across all monitored metrics.

A. *Data Availability*

The percentage of missing rows for each AWS station is shown in Fig. 1. The percentage of station data available for analysis is higher than 82% of all stations. Higher percentages of missing rows, roughly 40%, are displayed in the data for DAPODI AWS stations, which are exempt from QC. In the case of GPRS-based systems, communication failures and network problems are the leading causes of missing rows in AWS station data. or the system is off for a while because of a power problem.

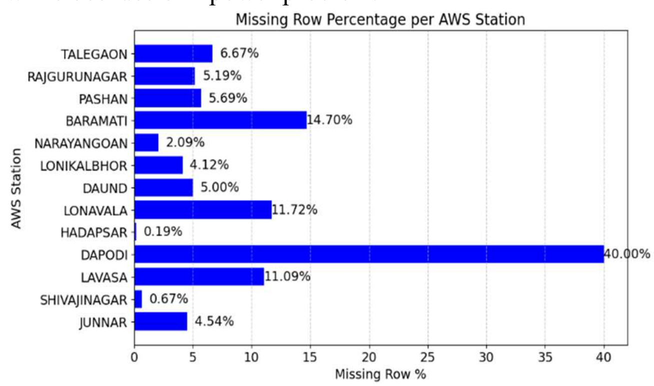

Fig. 1. Missing row per AWS stations

Fig. 2 shows the variables, including rainfall, temperature, and wind direction; the data availability across sites demonstrates great consistency, with Junnar and Lonikalbhor exhibiting the best performance. However, there are notable gaps at several places, particularly in the Wind Speed and Pressure data, such Dapodi, Hadapsar, and Talegaon. Pressure has the lowest availability overall (59.28%), indicating places where data coverage needs to be improved.

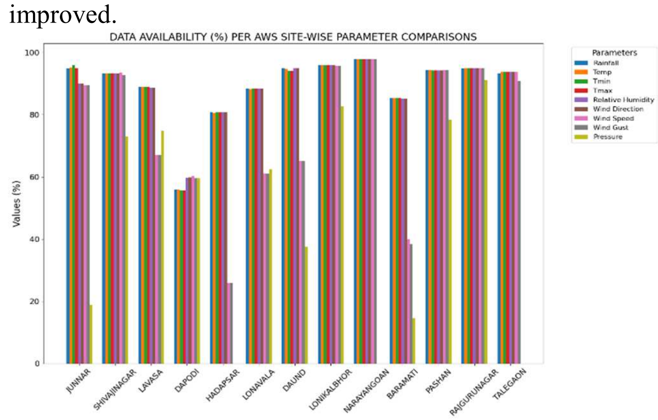

Fig. 2. Data Availability (%) per AWS site-wise comparison

B. *Range check*

Table IV shows the percentage of data that passed the range check. The available data, however, has met the specified range. Nonetheless, some station metrics, such as air temperature and relative humidity, are lower than others. This could happen if the sensor isn't working correctly. Even so, the range check has yielded positive outcomes. Most participants met the specified range, according to the range check results, which were generally positive. However, at several AWS sites, anomalies in temperature and relative humidity readings were noted. These anomalies are often the result of data encoding problems or malfunctioning sensors. Historical information and the climatological data of that specific geographic area are needed to correct this site. drift in sensor calibration, Wear or exposure to the environment can cause sensors to lose accuracy over time. Age, exposure to harsh environments, or physical injury can all cause the sensor to deteriorate or stop working. Inaccurate readings can result from a low or irregular power source. The range check may not work if the weather is genuinely extreme, such as a record-breaking high or low temperature.

TABLE IV. DATA PASSED RANGE CHECK (%)

Site Name	Parameters				
	Temp	***Relative Humidity***	***Wind Direction***	***Wind Speed***	***Pressure***
JUNNAR	99.24	68.944	93.94	93.37	94.41
SHIVAJINAGAR	99.97	86.46	99.98	99.98	78.16
LAVASA	98.24	75.86	75.42	99.6	96.21
DAPODI	93.15	90.13	99.62	99.28	99.28
HADAPSAR	99.87	94.34	99.26	99.21	-
LONAVALA	99.95	64.24	100	99.99	98.64
DAUND	99.65	84.315	100	68.59	100
LONIKALBHOR	100	76.85	100	99.75	86.15
NARAYANGOAN	99.98	85.08	100	99.98	-
BARAMATI	99.90	77.92	99.75	99.96	82.83
PASHAN	99.94	85.96	99.84	99.97	83.00
RAJGURUNAGAR	99.98	86.38	100	99.96	95.96
TALEGAON	99.99	86.93	99.97	99.20	-
Average	99.57	82.04	96.91	96.77	91.27

Checking the sensor for physical damage, contamination, or incorrect installation that could impair its effectiveness is the first step in fixing range check failures in autonomous weather station sensors. After that, verify the sensor's accuracy and performance in a controlled setting. Use the proper equipment to recalibrate the sensor if it's not functioning correctly or replace it if recalibration doesn't work. Ensure that the program settings and system configuration align with the region's realistic weather data expectations. Examine recent data records for odd patterns or anomalies that could lead to the underlying cause. Seek the manufacturer's technical assistance or consider replacing the problematic sensor if the problem continues. Keep a routine in place to avoid similar problems in the future. Maintain a program of cleaning and maintaining the sensors, doing routine calibration tests, and keeping an eye out for any early indications of sensor drift or failure to avoid similar problems in the future. Use backup sensors or cross-check data with neighboring stations to add redundancy to data collecting. With the temperature being the most stable (average of 99.57%) and relative humidity and pressure exhibiting more significant variability (82.04% and 91.27%), the table presents data pass rates for weather parameters across 13 sites. The humidity and wind speed pass rates were lower at locations like Lonavala and Daund.

C. *Consistency Check*

Table V demonstrates that the data is reliable and has a high degree of compliance. However, data reliability can be further improved by integrating extra checks for abnormal readings. Data consistency across five meteorological parameters: temperature, relative humidity, wind direction, wind speed, and rainfall at 13 sites is shown in Table V: Result of Consistency Check (%). With averages of 99.24% and 99.36%, respectively, rainfall and temperature exhibit outstanding stability, with several sites like Lavasa and Lonikalbhor achieving almost flawless marks. High dependability is also shown by Wind Direction and Wind Speed, which average 99.44% and 99.56%, respectively. At locations like Hadapsar and Daund, flawless scores have been recorded. Relative humidity, however, is the least consistent at 81.81%, with notable drops at Junnar (68.94%) and Lonavala (64.25%).

TABLE V. RESULT OF CONSISTENCY CHECK (%)

Site Name	Parameters				
	Rainfall	***Temp***	***Relative Humidity***	***Wind Direction***	***Wind Speed***
JUNNAR	98.88	99.24	68.94	93.94	95.96
SHIVAJINAGAR	99.99	99.97	86.48	99.98	99.98
LAVASA	100	100	75.86	99.61	99.61
DAPODI	93.01	93.16	90.13	99.62	99.82
HADAPSAR	99.87	99.87	94.34	100	100
LONAVALA	99.89	99.96	64.25	100	99.97
DAUND	99.94	99.66	84.32	100	100
LONIKALBHOR	100	100	76.85	100	100
NARAYANGOAN	98.98	99.98	85.09	100	100
BARAMATI	99.98	99.91	77.93	99.75	99.08
PASHAN	99.57	99.94	85.97	99.85	99.97
RAJGURUNAGAR	100	99.97	86.39	100	99.97
TALEGAON	100	99.98	86.93	100	99.92
Average	99.24	99.36	81.81	99.44	99.56

Although Relative Humidity indicates potential for improvement, the data is generally consistent across most factors and locations. Because of possible problems with sensor alignment, external influences, and local microclimate fluctuation, relative humidity exhibits the least consistency. Measurement inaccuracies may be made worse by locations like Lonavala and Junnar with varying circumstances. To enhance the overall quality of the data, step checks, persistence tests, and spatial consistency checks are essential. By detecting and fixing measurement problems, these checks assist in guaranteeing the accuracy of the data. These techniques greatly increase the data's dependability, producing more reliable outcomes. These inspections also aid in preserving uniformity over time and between various sites. Ultimately, they help improve the quality assurance of the data gathered, guaranteeing its analytical robustness.

Educating decisions on sensor replacements and maintenance requires routine calibration and methodical inspections. Putting pre-seasonal and post-seasonal maintenance into practice guarantees that sensors perform at their best all year. These preventative actions aid in the early detection of any possible problems, avoiding data errors brought on by sensor drift or malfunctions. Regular inspections also enable prompt replacement or repair of malfunctioning sensors, preserving the data's consistency and dependability. Ultimately, these procedures greatly increase the overall quality of the data, making weather observations more accurate and reliable.

V. CONCLUSION

To sum up, the AWS Lab data center efficiently carries out several checks to guarantee the precision and dependability of sensor data, including calibration, comparison, consistency, step, and range checks. The accuracy of the measurements can be further improved by evaluating dynamic thresholds using standard deviation, which is based on historical data and climatological normal. With an average of 99.57% across parameters, the data from Table IV demonstrates excellent performance in passing range checks. Impressive findings are also shown by consistency checks from Table V, particularly in the areas of rainfall, temperature, and wind speed, which show excellent consistency across all sites. The overall correctness and dependability of AWS data can be significantly increased by further honing these checks and using historical data. AWS systems exhibit robust data accuracy and consistency, validating their utility for dependable weather data collection and reinforcing their role in environmental monitoring and forecasting. Field comparison using the assistance of Traveling standards are secondary standards used to improve the quality of sensor data from ground stations at intervals of at least one year, six months, or at most quarterly intervals. QA/QC further enhances data reliability.

For high-quality data from the AWS/ARG station, routine calibration and maintenance are crucial. Quality assurance (QA) is vital to ensure data accuracy and reliability. The India Meteorological Department Standard Operating Procedures and WMO Guidelines will cover the on-site maintenance of hardware, software, sensors, electronics, and civil work. To improve the quality of data, the following maintenance process is necessary.

a) Preventive Maintenance: Preventive maintenance should be performed once a quarter. The manufacturer recommended inspections of automatic instruments and routine "housekeeping" at observation locations (such as mowing the lawn and cleaning exposed instrument surfaces).

b) Corrective Maintenance: should be promptly performed upon issue detection, with a system to identify equipment problems centrally, prioritizing repairs based on station type and defect severity within set timelines.

c) Adaptive Maintenance: After a few years, the availability of spare parts and the quick technological changes must be considered.

d) Calibration and comparison with reference standards.

e) For climatological purposes, maintaining history of instruments defects, exposure changes, and corrective actions in metadata and data is crucial.

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